Dmitry Tchoomak

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Profile

Lead software engineer with 18+ years of experience in media systems and video processing, including 14+ years combining technical leadership with hands-on development.
Specialization: media streaming, transcoding, multithreaded cross-platform services and applications, software architecture, CCTV/VSS, IPTV/OTT, DRM.
Delivered products from prototype to large-scale production.

Experience

Lead Software Engineer IntelComLine | 2026 - Present

- built a GPU transcoder filter on CUDA NVDEC/NVENC
- integrated a GPU video stabilization filter on OpenCV and CUDA NVDEC/NVENC
- integrated lightweight CPU analytics filters on OpenCV: motion detection, camera tampering
- added AV1 and WebRTC ingest and egress to the media server via a patched webrtc-java fork
- built a native desktop screen and camera streaming app for Astra Linux (X11/Fly), Windows and Kubuntu (Wayland) with hardware-accelerated encoding (H.264, H.265), self-healing RTMP and UVC/MIPI CSI-2 cameras

Head of Media Solutions Development Netris | 2012 - 2026

- built and delivered a passenger information video system for the Moscow Metro; in production since 2018
- led delivery of a cross-platform CCTV client for the "Safe City" state program; in production since 2020
- led development of IPTV/OTT services and STB clients; in production since 2013
- led development of a Digital Signage system based on Android TV
- worked closely with vendors and partners: Rockchip, Amlogic, Huawei, Hikvision, Dahua, Google/Widevine, Verimatrix

Software Engineer Smartlabs | 2008 - 2012

- developed video server subsystems (VoD, nPVR, TSTV) for an IPTV/OTT platform
- developed a restreamer-transcoder from various sources (YouTube, BBC, Blackmagic Design SDI)

Software Engineer Easy Trace Group | 2007 - 2008

- built components of a GIS vectorizer

Software Engineer BusinessInterSoft | 2003 - 2007

- built modules and tools for a native XML-based ODBMS

Skills

- C++20, Java, QML, SQL, XML/XSLT/XPath/DOM/SAX
- Qt 6, STL, Boost, Spring Boot, Netty, Android NDK, JNI, JavaCPP
- FFmpeg, CUDA, OpenCV, NVDEC/NVENC, Widevine
- MPEG-TS, MP4, FLV, H.264/AVC, H.265/HEVC, AV1, MP3, AAC, G.711, Opus
- RTSP, RTP/RTCP, RTMP, WebRTC, HLS, DASH, AMQP, RabbitMQ, TCP, UDP
- Clang, GDB, CMake, Maven, Bash, Git, GitLab, Jenkins, Docker, KVM/QEMU
- Linux (Wayland, X11), Windows, macOS, Android

Education

RSREA, Ryazan State Radio Engineering Academy Ryazan, RU | 2000 - 2005
Engineer's degree, Information Systems and Technologies

Certification

Widevine Modular DRM Kirkland, WA, US | 2015
Certified Widevine Implementation Partner

Widevine Classic VOD Packager Kirkland, WA, US | 2014
Certified Widevine Implementation Partner

Languages

Russian - native
English - B1, fluent technical reading and writing

Interests

WebAssembly, modding network hardware